THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 13 & 14: **Statistics, Probability** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of the first five even natural numbers is: 1

(a) 5 **(b)** 6 **(c)** 10 **(d)** 30

2. The upper limit of the class 15–25 is: 1

(a) 15 **(b)** 10 **(c)** 20 **(d)** 25

3. If the median is 15 and the mean is 16, the mode is: 1

(a) 13 **(b)** 14 **(c)** 18 **(d)** 17

4. A student states: "The cumulative frequency of the last class equals the total frequency." The correct response is: 1

(a) Only for odd n **(b)** Correct

(c) Incorrect **(d)** Cannot be determined

5. With assumed mean $a = 50$ and $\Sigma f_i d_i = 0$, Student A says the mean is 50; Student B says 0. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The probability of getting a 6 on a die is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** $\frac{1}{6}$

7. A letter is chosen from the word PROBABILITY. The probability that it is B is: 1

(a) $\frac{1}{11}$ **(b)** $\frac{2}{10}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{2}{11}$

8. The probability of drawing a face card from a deck of 52 cards is: 1

(a) $\frac{3}{13}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{13}$ **(d)** $\frac{4}{13}$

9. A student states: "If $P(E) = 0.6$, the event will surely happen." The correct response is: 1

(a) Incorrect — only $P(E) = 1$ means certain **(b)** Correct

(c) Correct above 0.5 **(d)** Cannot be determined

10. For P(sum 13) with two dice, Student A says 0; Student B says $\frac{1}{36}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of: 2

x	2	4	6	8
f	3	2	3	2

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12. Find the median class and modal class of: 2

Class	0–5	5–10	10–15	15–20	20–25
Frequency	4	7	15	10	4

13. A die is thrown. Find P(not a 6) and P(a number ≤ 2). 2

14. A bag has 4 red, 5 white and 6 black balls. Find the probability that a ball drawn is not black. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the mean by the assumed mean method: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	6	8	10	9	7

16. A carton of 50 eggs has 5 rotten eggs. 3
(i) Find the probability that an egg drawn is good.
(ii) If 3 good eggs are first removed, find the probability that the next egg is rotten.

17. Find the median of: 3

Class	10–20	20–30	30–40	40–50	50–60
Frequency	5	9	14	8	4

OR

Find the mode of: 3

Class	0–20	20–40	40–60	60–80	80–100
Frequency	5	8	13	7	7

18. All four aces are removed from a deck. One card is drawn from the rest. Find P(king), P(red card) and P(ace). 3

OR

A number is chosen from 1 to 30. Find the probability that it is a multiple of 4 or 6. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the median and the mode of daily wages: 5

Wages (₹)	100–150	150–200	200–250	250–300	300–350
Workers	6	10	15	12	7

OR

The mean pocket allowance is ₹18. Find the missing frequency f, and state the modal class. 5

Allowance (₹)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Children	7	6	9	13	f	5	4

20. **Case study:** In a class of 40, 15 students like cricket only, 10 like football only and the rest like neither. A student is chosen at random. Find 5
(i) P(likes cricket)
(ii) P(likes neither)
(iii) P(likes cricket or football).
(iv) A student says P(cricket) + P(football) = 1. Comment.

OR

A red die and a blue die are thrown. Find the probability that 5
(i) the red die shows a larger number
(ii) the blue die shows 6
(iii) the sum is at most 5.

END OF QUESTION PAPER — Total: 40 Marks